

THE MURMURATION OF A CM QUADRATIC TWIST FAMILY: EXACT REDUCTION, CLOSED-FORM ROOT NUMBERS, THE MEASURED GRÖSSENCHARACTER WAVEFORM, AND THE VERIFIED POISSON ENGINE

ABSTRACT. For the quadratic twist family of $E_0 = 27a3$ ($y^2 + y = x^3$, CM by $\mathbb{Q}(\sqrt{-3})$) we prove an exact reduction of the murmuration to elementary arithmetic: the root numbers of the entire family admit a two-line closed form (via a CM self-twist argument eliminating the local analysis at 3; verified 153/153), the traces are given by $4p = L^2 + 27M^2$, and the density factors through a single character sum over fundamental discriminants. The resulting pipeline is certified against exact computation to five decimal places and requires no elliptic-curve arithmetic. We then reduce the family waveform to a single-object function: the murmuration $\overline{M}(s)$, $s = p/N_D$, of the fixed Hecke Grössencharacter of $\mathbb{Q}(\sqrt{-3})$ under quadratic twist. Empirically $\overline{M}$ is universal across disjoint discriminant ranges (correlation +0.945) and is resolved at $5\text{--}16\sigma$: a principal peak +7.4 near $s \approx 0.03$, oscillation with a deep trough -8.3 just below the conductor scale and a rebound +7.3 just above it, decaying to the explicit-formula constant $C_0 = \frac{1}{2}\overline{w} - \overline{w}r = +0.4085$ (confirmed exactly from rank parities). The exact transform of $\overline{M}$ reproduces the directly measured family profile on its resolved bins. The derivation of $\overline{M}$ in closed form is located within the published murmuration framework: the Dirichlet-character (GL(1)) densities of Lee–Oliver–Pozdnyakov, the explicit-formula mechanism for quadratic families, and Sarnak’s conductor-growth framework; our object—the quadratic twist family of a Grössencharacter over an imaginary quadratic field—appears to be new, and every arithmetic ingredient of its Zubrilina-type computation is supplied here in verified closed form.

1. THE EXACT REDUCTION

Let $E_D = E_0^{(D)}$ for fundamental discriminants D , and $F(X) = \{0 < |D| \leq X\}$. All results below are machine-certified; scripts are available with this note.

Proposition 1 (Reduction). *For p of good reduction for the family, $\frac{1}{|F|} \sum_D w(E_D) \frac{a_p(E_D)}{\sqrt{p}} = \frac{a_p(E_0)}{\sqrt{p}} S(p)$ with $S(p) = \frac{1}{|F|} \sum_D w(E_D) \left(\frac{D}{p}\right)$. Verified over 57,681 pairs; the two sides, computed independently, agree to 6.9×10^{-18} .*

Proposition 2 (Closed-form root numbers). *For fundamental D , $|D| \leq 250$: $w(E_D) = \left(\frac{D}{-27}\right)$ if $\gcd(D, 3) = 1$, and $w(E_D) = \left(\frac{-D/3}{-27}\right)$ if $3 \mid D$. Proof for $3 \mid D$: the self-twist $E_0^{(-3)}$ has $a_p(E_0^{(-3)}) = \chi_{-3}(p)a_p(E_0) = a_p(E_0)$ at every good prime, additive reduction at 3 on both sides, and equal conductor; hence identical L -functions and equal functional-equation sign, so with $D = (-3)m$ (prime-discriminant factorization), $w(E_D) = w(E_0^{(m)})$, reducing to the coprime case. Verified 114/114 (coprime, including even D) and 39/39 ($3 \mid D$).*

Corollary 3 (Curve-free computability). *With $a_p(E_0) = 0$ at inert p and, at split p , the root of $4p = L^2 + 27M^2$ congruent to 2 mod 3, the murmuration is elementary arithmetic. The pipeline reproduces exact computation to five decimals ($+0.03011 = +0.03011$), and $S(p)$ factors through $T(Y; p) = \sum_{\text{fund } m, 3 \nmid m, |m| \leq Y} \left(\frac{m}{-27p}\right)$.*

Date: July 18, 2026.

2. REDUCTION TO A SINGLE GRÖSSENCHARACTER; THE MEASURED WAVEFORM

Every E_D has CM by $\mathbb{Q}(\sqrt{-3})$: its L -function is that of a Hecke Grössencharacter. Define the *single-object murmuration*

$$\overline{M}(s) = \langle w(E_D) a_p(E_D) \rangle_{p=sN_D}, \quad N_D = \begin{cases} 27D^2, & 3 \nmid D, \\ 3D^2, & 3 \mid D, \end{cases}$$

each twist in its own conductor-scaled variable. The family profile is the exact transform $g(x) = \langle \overline{M}(x \cdot 27X^2/N_D) \rangle_{D \in F(X)}$.

Observation 4 (Universality and the waveform). Superposing all 153 twists (1,370,136 samples, inverse-variance weighted), $\overline{M}$ measured from $|D| \leq 100$ and from $|D| > 100$ agree with correlation +0.945: the single-object waveform exists. Its measured values:

s -bin	$\overline{M}(s) \pm \text{SE}$	feature
[0, 0.02)	+0.01 $\pm$ 0.13	flat onset
[0.02, 0.05)	+7.40 $\pm$ 0.46	principal peak
[0.05, 0.10)	+4.10 $\pm$ 0.57	
[0.10, 0.20)	−4.17 $\pm$ 0.60	first sign change
[0.20, 0.35)	−1.61 $\pm$ 0.72	
[0.35, 0.50)	+5.60 $\pm$ 0.97	second peak
[0.50, 0.75)	+0.20 $\pm$ 1.02	
[0.75, 1.00)	−8.31 $\pm$ 1.26	pre-conductor trough
[1.00, 1.50)	+7.33 $\pm$ 1.12	post-conductor rebound
[1.50, 2.50)	+0.01 $\pm$ 1.10	
[2.50, 4.00)	+1.41 $\pm$ 1.32	decay toward C_0

The structure at the conductor scale (trough below $s = 1$, rebound above) is the functional-equation transition made visible. Reproduced deterministically on independent hardware from an 80-line script using only the closed forms.

Observation 5 (Transform closure). The measured $\overline{M}$, pushed through the exact transform, predicts the directly measured family profile: on its two resolved bins, predicted +1.79, +1.91 versus measured +2.19 $\pm$ 0.42, +1.98 $\pm$ 0.57; 13 of 15 bins agree within 1σ . The apparent irresolution of the family profile beyond its front is thereby explained as *phase-mixing*: each twist oscillates at its own conductor scale, and the family variable averages $\overline{M}$ over a wide range of its argument.

Observation 6 (The constant). $C_0 = \frac{1}{2}\overline{w} - \overline{w\overline{r}}$ with exact ranks gives +0.4085 ($\overline{w} = -7/153$, $\overline{w\overline{r}} = -66/153$), derived from rank parities in advance and reproduced exactly; $C_0 \rightarrow \frac{1}{2}$ under Goldfeld and parity.

3. THE POISSON ENGINE, VERIFIED

The remaining computation is now implemented and machine-verified. The Hecke character of E_0 has the exact lattice description $\psi_0(\mathfrak{a}) = \lambda$, the unique generator with $\lambda \equiv 1 \pmod{3}$; equivalently, for $\gcd(n, 3) = 1$,

$$a_n(E_0) = \sum_{\substack{a \equiv 1, \, b \equiv 0 \pmod{3} \\ a^2 - ab + b^2 = n}} (a + b\omega),$$

verified exactly for all $n \leq 2000$ (zero mismatches), and the twisted sums $\sum_n \chi_D(n) a_n g(n/P)$ equal the corresponding lattice sums to machine precision. Two-dimensional Poisson summation over $\mathbb{Z}[\omega]$, with the Gaussian-times-linear envelope transformed in closed form (checked against numerical integration at 10^{-9}), then gives, for $P = sN_D$,

$$\sum_n \chi_D(n) a_n(E_0) e^{-2\pi n/(sN_D)} = s^2 \sum_{\mu \in \mathbb{Z}^2} \gamma_\mu(D) e^{-2\pi s Q^*(\mu)}, \quad Q^*(\mu) = \mu_1^2 + \mu_1 \mu_2 + \mu_2^2,$$

with $\gamma_\mu(D)$ explicit finite Gauss sums (the discrete Fourier coefficients of the periodic datum: primarity mod 3 and $\chi_D \circ Q \bmod |D|$). The identity is verified at machine precision (1.3×10^{-14} at $s = 1$; 3.5×10^{-15} at $s = 0.05$) for $3 \nmid D$ ($D = 5$, $M = 3|D|$) and for $3 \mid D$ ($D = -24$, $M = |D|$); in both cases the conductor normalization makes the dual exponent *exactly* $2\pi s Q^*(\mu)$ —deriving, rather than assuming, that the murmuration variable is $s = p/N$. The dual support is sharply localized on hexagonal shells: at $s = 1$ the first shell ($Q^* = 1$, six vectors) carries 100% of the sum; three shells at $s = 0.3$; ten at $s = 0.05$ —an $O(1/s)$ term count matching the measured waveform’s simple conductor-scale structure and rich front. The truncation $Q^* \leq 30$ reproduces the exact sum at relative error 10^{-15} for $s \geq 0.1$.

Step (i) is now closed in the generic case. For odd fundamental D with $3 \nmid D$ (period $M = 3|D|$, $\alpha \equiv |D|^{-1} \bmod 3$):

$$\gamma_\mu(D) = \frac{1}{9|D|} \omega_3^{\alpha\mu_1} \chi_{-3}(|D|) \chi_D(-3) \chi_D(Q^*(\mu)),$$

derived by completing the square against the norm form (whose Gauss sum is $\chi_{-3}(q)q$, independent of the multiplier—the hexagonal miracle) with the $\tau(\chi_D)$ factors cancelling identically. Verified at 10^{-17} against brute discrete transforms across six discriminants (224 dual vectors each), and the resulting fully explicit dual formula—kronecker symbols and exponentials only—reproduces the exact lattice sums at 10^{-11} – 10^{-13} . Since $w(E_D) = \chi_D(-27) = \chi_D(-3)$ for these D , the root-number weight cancels the Gauss phase exactly:

$$w_D \gamma_\mu(D) = \frac{\chi_{-3}(|D|)}{9|D|} \omega_3^{\alpha\mu_1} \chi_D(Q^*(\mu)) :$$

the murmuration weight is precisely the conjugating phase of the dual expansion—the structural reason this family murmurs coherently. For the excluded cases the same skeleton holds numerically: $|\gamma| \cdot 9|D| = 1$ exactly for even D and $= 3$ for $3 \mid D$ (ramification enhancement), with the $\chi_D(Q^*)$ -proportionality exact in all tested vectors; only the unit phases at 2 and 3 remain as classical case-work.

Step (ii) is now closed at the main-term level for the generic subfamily. The character sums $T_\mu(X) = \sum_D D^2 \chi_{-3}(|D|) \omega_3^{\alpha\mu_1} \chi_D(Q^*(\mu))$ have main terms exactly at square shells $Q^*(\mu) = k^2$ with $\mu_1 \not\equiv 0 \bmod 3$ (verified: alive vectors converge at X^3 -order; dead vectors, dead shells such as $Q^* = 9$, and non-square shells decay — within the single shell $Q^* = 1$, $\mu = (1, 0)$ converges while $\mu = (0, 1)$ dies). Pairing each shell in $\pm\mu$, the real part of T_μ cancels identically—the equidistribution error drops out exactly—and the shell geometry evaluates to $c^T A_0^{-1}(2k, -k) = 2k$, giving the closed-form main term of the w -averaged Gaussian-smoothed family sum:

$$\lim_{X \rightarrow \infty} \frac{A(X, s)}{\sum_{|D| \leq X} D^2} = 27 s^2 \sum_{k \geq 1} \chi_{-3}(k) k e^{-2\pi s k^2}$$

(up to Euler-factor corrections for k with prime factors ≥ 5 and extra Lösschian representations from $k^2 = 49$ onward, both entering at $e^{-50\pi s}$). The formula reproduces the computed family averages to all printed digits at $s = 0.5$, 1 ($+0.29165$, $+0.05042$), and at small s the non-square residual is exhibited decaying in X ($-5.7 \times 10^{-4} \rightarrow -6.0 \times 10^{-5}$ over $X = 400 \rightarrow 3200$ at

$s = 0.05$). The main term is a harmonic theta series with character χ_{-3} —an η^3 -adjacent object whose modular identification we flag but do not assert.

What remains: (iii) unsmoothing (Gaussian to sharp windows) and the coefficient-to-prime conversion, transforming the theta main term into the sharp waveform; and the extension of steps (i)–(ii) to even D and $3 \mid D$ (unit phases; the structure is already verified numerically). Every analytic ingredient is now explicit.

4. THE REMAINING STEP, LOCATED

Closed-form murmuration densities for $\mathrm{GL}(1)$ over $\mathbb{Q}$ are established: Lee–Oliver–Pozdnyakov compute the densities for two families of Dirichlet characters, the real-character density exhibiting Zubrilina-type universality and interpolating the symplectic one-level phase transition [LOP]. The explicit-formula mechanism for quadratic families is given in [BEF], and Sarnak’s framework ties the density’s form to the conductor growth of the family [Sar, Zub]. The object measured here—the quadratic twist family of a fixed Grössencharacter of an imaginary quadratic field, conductor growth $N_D \asymp D^2$ —does not appear in these works. Its closed-form derivation is the application of the $\mathrm{GL}(1)$ template to a fully explicit input: the trace $a_p = \pi + \bar{\pi}$ is a lattice sum over $\mathbb{Z}[\omega]$, the weights are the closed forms of Proposition 2, and Poisson summation over the lattice is the remaining computation. Observation 4 is the answer key any candidate density must reproduce.

Remark 7 (Scope). Proved and certified: Propositions 1–2, Corollary 3. Measured, with controls (universality across disjoint subfamilies, transform closure, deterministic reproduction): Observations 4–6. Not done: the closed-form evaluation of $\overline{M}$; it is posed with all ingredients supplied. Novelty of the measured object is asserted only as absence from the works cited, pending a full literature sweep. We claim nothing further.

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